

COURSE: CS 201 — Data Structures and Algorithms II

Credits: 4 | Department: Computer Science

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PREREQUISITES: CS 102 (minimum grade C)

CO-REQUISITES: None

OFFERED: Fall, Spring

DESCRIPTION:

Advanced data structures and algorithm design. Topics include advanced graph algorithms (Dijkstra's, Bellman-Ford, Floyd-Warshall, minimum spanning trees), dynamic programming, greedy algorithms, amortized analysis, balanced BSTs (Red-Black trees, B-trees), and NP-completeness introduction. Students work in groups on a semester-long algorithmic problem-solving project.

MINIMUM GRADE TO SATISFY PREREQUISITE FOR DOWNSTREAM COURSES: C (required for CS 220, CS 230, CS 240, CS 250, CS 260, CS 301)

GRADING: Programming assignments 45%, Midterm 20%, Final 25%, Group project 10%.